



Operations Research

Prof. Dr. Martin Bichler,
Halil Bayrak, Janik Bürgermeister, Teodora Dobos,
Jan Höhener, Abheek Ghosh

Solution to the Training Exercises, Sheet 5

Training Exercises

Exercise T5.1. *Modelling Tricks - OR*

Problem

Let $x, y \geq 0$ and $x, y \in \mathbb{Z}$. How can we formulate conditions such that either $x + y \leq 3$ or $2x + 5y \leq 12$ (or both) always hold? How to model it so that exactly one of these conditions holds (XOR)?

Solution

Solution 1 for OR:

Define two auxiliary variables $z_1, z_2 \in \{0, 1\}$ and a large number M to model the implication using the following constraints:

$$x + y - 3 \leq Mz_1 \quad (1)$$

$$2x + 5y - 12 \leq Mz_2 \quad (2)$$

$$z_1 + z_2 \leq 1 \quad (3)$$

$$z_1, z_2 \in \{0, 1\} \quad (4)$$

We should ensure that M is large enough so that, for example, $x + y - 3$ is always at most as large as M (e.g., guaranteed by other constraints). M is not a variable, but a fixed number. This method is useful to define an OR operation even over a larger number of constraints.

Solution 2 for OR:

Here we need only one auxiliary variable $z \in \{0, 1\}$ and the following constraints:

$$x + y - 3 \leq Mz \quad (5)$$

$$2x + 5y - 12 \leq M(1 - z) \quad (6)$$

$$z \in \{0, 1\} \quad (7)$$

Solution 3 for OR (and XOR):

It is also possible to define a separate binary variable for each constraint, which indicates whether the constraint is fulfilled or not. Here, however, one must pay attention to the fact that the negation of a constraint often contains a $>$ or $<$ and thus may not be represented so simply in an IP. In the case of integer variables (as here), however, one can often model them again using $\leq, \geq$.

$$x + y - 3 \leq M(1 - z_1) \quad (8)$$

$$4 - x - y \leq Mz_1 \quad (9)$$

$$2x + 5y - 12 \leq M(1 - z_2) \quad (10)$$

$$13 - 2x - 5y \leq Mz_2 \quad (11)$$

$$z_1 + z_2 \geq 1 \quad (12)$$

$$z_1, z_2 \in \{0, 1\} \quad (13)$$

This option also allows us in a very simple way to define other logical combinations of constraints. For XOR, for example, only one of the constraints may be valid. Then we replace the corresponding condition in the above model by $z_1 + z_2 = 1$.

Exercise T5.2. *Caffeine*

Problem

Maxime is often very tired at work. She describes her tiredness by a *tiredness level*. This is a natural number from the set $\{0, \dots, 20\}$, where 20 means very tired, and 0 means wide awake. To counteract her tiredness, she can buy various caffeinated drinks:

Drink	Decrease in tiredness	Price (euro)
Coffee	11	3
Black Tea	7	2
Green Tee	3	1

Maxime works 8 hours each Monday through Friday. In each working hour, she can choose to consume none or exactly one of the drinks. If she chooses to drink none, her tiredness level increases by 2 during that hour. If she chooses to drink one of the drinks, her tiredness level decreases by the corresponding value in the *Decrease in tiredness* column. The following additional conditions must be met:

- The tiredness level on Mondays in the first hour of work is 10.
- For any other workday, if any of the drinks were consumed during the 6th, 7th, or 8th hour of work the day before, Maxime starts the day with a fatigue level of 13, otherwise 10.
- Maxime can't spend more than 30 euros a week on drinks.
- Maxime can't drink a beverage that would make her tiredness level drop below 0.
- She must drink one of the three drinks once her fatigue level is greater than 18.

- Maxime doesn't want to drink coffee at least one day a week.

Maxime wants to minimize the sum over all tiredness levels of all hours worked in a week. Formulate this problem as IP without solving it.

Solution

Let $I = \{1, \dots, 8\}$ be the set of hours in the day, $D = \{1, \dots, 5\}$ be the set of working days, where 1 corresponds to Monday, 5 to Friday. We introduce variables $c_{id}, b_{id}, g_{id} \in \{0, 1\}$ that describe whether coffee (c), black tea (b), or green tea (g) is consumed in hour i on day d . Since only a maximum of one beverage may be consumed in each hour, we obtain the first constraint

$$c_{id} + b_{id} + g_{id} \leq 1 \quad \forall i \in I, d \in D$$

Further, we define variables $t_{id} \in \mathbb{Z}$ that describe the tiredness at time i, d .

We now come to the initial tiredness levels in each day. Monday morning, tiredness level is 10:

$$t_{11} = 10$$

For all other days, the tiredness depends on whether something was consumed in the last hours of the previous day. For this, we denote by $A_d \in \{0, 1\}$ whether something was consumed in the last 3 hours of day d :

$$3A_d \geq \sum_{i=6}^8 c_{id} + b_{id} + g_{id} \quad \forall d \in D$$

The start value on the next day is defined as

$$t_{1d} = 10 + 3A_{d-1} \quad \forall d \in \{2, \dots, 5\}$$

Further, the consumption of drinks affects the tiredness level:

$$t_{id} = t_{(i-1)d} + 2 - 13c_{id} - 9b_{id} - 5g_{id} \quad \forall d \in D, i \in \{2, \dots, 8\}$$

Maxime must not exceed her budget of 30 euros:

$$\sum_{i \in I, d \in D} 3c_{id} + 2b_{id} + g_{id} \leq 30$$

In addition, her tiredness level must never fall below 0 or exceed 20:

$$t_{id} \geq 0 \quad \forall i \in I, d \in D$$

$$t_{id} \leq 20 \quad \forall i \in I, d \in D$$

Maxime must drink one of the three drinks as soon as her tiredness level is greater than 18. Note that if the tiredness level is 19 or higher, and Maxime does not drink, then the tiredness level on the next day would increase by two and would therefore be at least 21. To avoid this, we set the upper bound of 20. Maxime does not want to drink coffee on at least one day. We use $C_d \in \{0, 1\}$ to describe whether she drinks coffee on day d :

$$8C_d \geq \sum_{i \in I} c_{id} \quad \forall d \in D$$

Then the following must hold:

$$\sum_{d \in D} C_d \leq 4.$$

Maxime would like to minimize the sum of the tiredness levels, i.e.,

$$\min \sum_{i \in I} \sum_{d \in D} t_{id}$$

Exercise T5.3. Exam Preparation

Problem

The teaching assistants Georgina and Maxime are planning the exam for the lecture “Molecular Biology for Information Systems”. They have created a collection of 20 potential *tasks* $a \in A := \{1, \dots, 20\}$ and assigned to each of these tasks a *complexity* $k_a \in \{1, \dots, 10\}$ as well as an *estimated completion time in minutes* $d_a \in \mathbb{N}$. Furthermore, the tasks can be classified into two *types*: calculus tasks and multiple choice tasks. The parameter $t_a \in \{0, 1\}$ describes the type of task $a \in A$, where $t_a = 0$ means that a is a calculus task, while $t_a = 1$ means that a is a multiple choice task. The exam may consist of *at most* 12 tasks.

Georgina and Maxime would like to leave the design of what they consider the most optimal exam to Gurobi. They therefore introduce binary variables $x_{as} \in \{0, 1\}$ for $a \in A$ and $s \in S := \{1, \dots, 12\}$, where $x_{as} = 1$ indicates that task a is on the s -th page of the exam. The exam must satisfy several constraints. Formulate the following constraints as linear (in)equalities. You *may* introduce additional variables. If you do, state the intuitive meaning of the variables.

- a) There may be no more than one task on each page $s \in S$ of the exam.
- b) Each task may be in the exam no more than once.
- c) The total complexity of the exam is the sum of the complexities of the tasks it contains and must be at least 40.
- d) The estimated time required to complete the exam must not exceed 115 minutes.
- e) No three tasks of the same type may follow each other consecutively.
- f) The exam should not have a blank page: If there is no task on the s -th page of the exam, then there is no task on all subsequent pages $s' > s$.
- g) If the exam contains an easy task with complexity $k_a \leq 2$, then it must also contain at least one difficult task with complexity $k_a \geq 8$.
- h) Georgina and Maxime have two objectives for the exam: (1) the exam should have a complexity of 60 and (2) the exam should have a completion time of 95 minutes. The complexity objective is twice as important to them as the completion time objective. Formulate an objective function that aggregates both objectives. Introduce additional variables and constraints if necessary.

Solution

a) No more than one task per page

$$\sum_{a \in A} x_{as} \leq 1 \quad \forall s \in S$$

b) Each task not more than once

$$\sum_{s \in S} x_{as} \leq 1 \quad \forall a \in A$$

c) Lower bound for total complexity of the exam

$$40 \leq \sum_{a \in A, s \in S} x_{as} k_a$$

d) Upper bound for estimated time to complete the exam

$$\sum_{a \in A, s \in S} x_{as} d_a \leq 115$$

e) No three tasks of the same type immediately after each other

$$\sum_{s=r}^{r+2} \sum_{a \in A} x_{as} t_a \leq 2 \quad \forall r \in \{1, \dots, 10\}$$

$$\sum_{s=r}^{r+2} \sum_{a \in A} x_{as} (1 - t_a) \leq 2 \quad \forall r \in \{1, \dots, 10\}$$

Note that both constraints are needed due to the values that t_a can take (0 or 1).

f) No blank pages in between

$$\sum_{a \in A} x_{as} \geq \sum_{a \in A} x_{a(s+1)} \quad \forall s \in \{1, \dots, 11\}$$

g) If the exam contains an easy task with $k_a \leq 2$, it has to contain a hard task with $k_a \geq 8$:

We introduce the binary variable $y \in \{0, 1\}$ such that “ $\exists$ Task with complexity $\leq 2 \Rightarrow y = 1$ ”:

$$My \geq \sum_{a \in A} x_{as} (3 - k_a) \quad \forall s \in S$$

Furthermore, we introduce the binary variable $z_s \in \{0, 1\}$ with $s \in S$ such that “Page s in the exam has a task with complexity $\leq 7 \Rightarrow z_s = 0$ ”:

$$8z_s \leq \sum_{a \in A} x_{as} k_a \quad \forall s \in S$$

Then the following must hold

$$\sum_{s \in S} z_s \geq y$$

h) Target complexity: 60. Target working time: 95. Introduce variables $\delta^k \in \mathbb{R}$ and $\delta^d \in \mathbb{R}$, indicating the deviations from 60 and 95, respectively:

$$\begin{aligned}\sum_{a \in A, s \in S} x_{as} k_a + \delta^k &= 60 \\ \sum_{a \in A, s \in S} x_{as} d_a + \delta^d &= 95\end{aligned}$$

Let $\Delta^k, \Delta^d \in \mathbb{R}$ be the absolute values of δ^k and δ^d , respectively:

$$\begin{aligned}\Delta^k &\geq \delta^k \\ \Delta^k &\geq -\delta^k \\ \Delta^d &\geq \delta^d \\ \Delta^d &\geq -\delta^d\end{aligned}$$

Then, the joint objective function is:

$$\min 2\Delta^k + \Delta^d$$

Self-study Exercises

Exercise S5.1. *Logic*

Problem

Using integer linear programs, it is easy to impose logical conditions on the variables. In the following, $A \in \{0, 1\}$ and $B \in \{0, 1\}$ denote binary variables, where as usual 1 corresponds to *true* and 0 to *false*.

- a) The binary variable $Z_{\neg A} \in \{0, 1\}$ is said to express the negation of A , so $Z_{\neg A} = 1 \Leftrightarrow A = 0$. Express this using linear (in-)equalities.
- b) The binary variable $Z_{A \vee B} \in \{0, 1\}$ is said to express the disjunction of A and B , i.e. $Z_{A \vee B} = 1 \Leftrightarrow A \vee B = 1$. Express this using linear (in-)equalities.
- c) Do the same for the conjunction $A \wedge B$ and the implication $A \rightarrow B$.

Also, visualize the feasible set in each case.

Now let $P \in [0, 100]$ be a real-valued variable.

- d) For the binary variable Z let $P < 10 \Rightarrow Z = 1$ and $P > 10 \Rightarrow Z = 0$ hold. Again, express this using linear (in-)equalities.
- e) Is it possible to define a variable Z with $Z = 1 \Leftrightarrow P \leq 10$?
Hint: What would the corresponding polytope have to look like?

Solution

- a) The equation we are looking for is $Z_{\neg A} = 1 - A$
- b) For $A \vee B$, for example, the inequalities

$$\begin{aligned} Z_{A \vee B} &\geq A \\ Z_{A \vee B} &\geq B \\ Z_{A \vee B} &\leq A + B \end{aligned}$$

can be used: If A or B is true, then one of the first two inequalities enforces that $Z_{A \vee B}$ is also true. On the other hand, if A and B are both false, then $Z_{A \vee B}$ must also be false because of the third inequality.

- c) For $A \wedge B$:

$$\begin{aligned} Z_{A \wedge B} &\leq A \\ Z_{A \wedge B} &\leq B \\ Z_{A \wedge B} &\geq A + B - 1 \end{aligned}$$

For $A \Rightarrow B$:

$$\begin{aligned} Z_{A \Rightarrow B} &\geq 1 - A \\ Z_{A \Rightarrow B} &\geq B \\ Z_{A \Rightarrow B} &\leq 1 + 2B - A \end{aligned}$$

$A \Rightarrow B$ is not satisfied only if A is true and B is false. Then the third inequality enforces $Z_{A \Rightarrow B} = 0$. In all other cases, the first two inequalities ensure $Z_{A \Rightarrow B} = 1$.

d) One possible solution is

$$Z \geq 1 - \frac{P}{10}$$
$$Z \leq \frac{10}{9} - \frac{P}{90}$$

The easiest way to see this is by visualizing the feasible set.

e) Not possible, because the feasible set is not closed: For every $\epsilon > 0$ $Z = 0, P = 10 + \epsilon$ is feasible, but the limit $Z = 0, P = 10$ is not feasible. Therefore, the feasible set cannot be represented as a polytope, so in particular it cannot be represented by linear inequalities.

Exercise S5.2. Party Planning

Problem

To celebrate your successful final exam in OR, you spontaneously decide to throw a party. To do this, you want to buy drinks from the following selection at the nearby supermarket. Note that you can only buy whole bottles at the supermarket!

Beverage	Price per bottle	Volume per bottle
Sparkling water	0,5 €	1 l
Orange juice	0,7 €	1 l
Vodka	7 €	0.7 l
Beer	0,8 €	0.5 l

You decide to meet the following constraints:

- You want to buy at least 10 liters of non-alcoholic drinks (sparkling water or orange juice).
- You want to buy at least 10 bottles of beer and one bottle of vodka.
- You want to buy at least 40 bottles of beer or at least 5 bottles of vodka. (OR)
- When you buy vodka, in addition to the orange juice purchased for non-alcoholic use, you want to buy another 2 bottles of orange juice per bottle of vodka.
- For weight reasons, you want to buy at most 30 liters of beverages. However, the supermarket offers you free home delivery for purchases of 50 € or more. In this case, you can buy as much as you want without having to worry about the weight.

Formulate a mixed-integer linear program to buy drinks for your party as cheap as possible while satisfying this set of constraints.

Solution

Variables

$x_M \in \mathbb{N}_0$	Purchased bottles of sparkling water
$x_O \in \mathbb{N}_0$	Purchased bottles of orange juice
$x_W \in \mathbb{N}_0$	Purchased bottles of vodka
$x_B \in \mathbb{N}_0$	Purchased bottles of beer
$x_O^{AF} \in \mathbb{N}_0$	Purchased bottles of orange juice (for non-alcoholic use)
$x_O^M \in \mathbb{N}_0$	Purchased bottles of orange juice (for mixing with vodka)
$z \in \{0, 1\}$	Auxiliary variable for the third constraint
$l \in \{0, 1\}$	Auxiliary variable for home delivery

Model

$$\min \quad 0.5x_M + 0.7x_O + 7x_W + 0.8x_B \quad (1)$$

$$\text{s.t.} \quad x_O^{AF} + x_M \geq 10 \quad (2)$$

$$x_B \geq 10 \quad (3)$$

$$x_W \geq 1 \quad (4)$$

$$40 - x_B \leq Mz \quad (5)$$

$$5 - x_W \leq M(1 - z) \quad (6)$$

$$x_O^{AF} + x_O^M = x_O \quad (7)$$

$$x_O^M \geq 2x_W \quad (8)$$

$$50 - (0.5x_M + 0.7x_O + 7x_W + 0.8x_B) \leq M(1 - l) \quad (9)$$

$$x_M + x_O + 0.7x_W + 0.5x_B \leq 30 + Ml \quad (10)$$

$$x_M, x_O, x_W, x_B, x_O^{AF}, x_O^M \in \mathbb{N}_0 \quad (11)$$

$$z, l \in \{0, 1\} \quad (12)$$

Exercise S5.3. Dutch Petroleum

Problem

Dutch Petroleum (DP) produces two different types of fuel and requires two different types of crude oil for this. The following quality requirements apply.

- Fuel 1: Minimum crude oil type 1: 50%
- Fuel 2: Minimum crude oil type 1: 60%

DP currently holds 10 000l of crude oil 1 and 20 000l of crude oil 2. DP is able to purchase a maximum of 30 000l of crude oil type 1 at the following prices, whereby DP receives a volume discount for large purchases: The price is 1.8€/l if less than 10 000l is purchased and 1.6€/l if more than 10 000l is purchased.

Fuel 1 can be sold for 1.4€/l, fuel 2 even for 1.5€/l.

Formulate an integer linear optimization model to solve this problem, with the aim of optimizing the profit (i.e., revenue - costs).

Solution

The problem is a classic mixed problem, combined with an either-or condition. We need the variables

Variables

- k = The amount purchased from crude oil 1
- x_{ij} = The amount of oil i used for fuel j ($i, j = 1, 2$)

Model

The model, without considering the non-linearity of the cost function, could look as follows.

$$\begin{aligned}
 \max z = & 1.4x_{11} + 1.4x_{21} + 1.5x_{12} + 1.5x_{22} - c(k) \\
 \text{s.t. } & x_{11} + x_{12} \leq 10\,000 + k \\
 & x_{21} + x_{22} \leq 20\,000 \\
 & 0.5x_{21} - 0.5x_{11} \leq 0 \\
 & 0.6x_{22} - 0.4x_{12} \leq 0 \\
 & k \leq 30\,000 \\
 & k \geq 0 \\
 & x_{ij} \geq 0 \quad \forall i \in \{1, 2\}, j \in \{1, 2\}
 \end{aligned}$$

option 1: One way to model the costs is to formulate them with an either-or link. We formulate 2 cost functions here, whereby the more favorable cost function may only be used if the decrease is greater than or equal to 10 000

$$\begin{aligned}
 \max z = & 1.4x_{11} + 1.4x_{21} + 1.5x_{12} + 1.5x_{22} - 1.8y_1 - 1.6y_2 \\
 \text{s.t. } & x_{11} + x_{12} \leq 10\,000 + k \\
 & x_{21} + x_{22} \leq 20\,000 \\
 & 0.5x_{21} - 0.5x_{11} \leq 0 \\
 & 0.6x_{22} - 0.4x_{12} \leq 0 \\
 & k = y_1 + y_2 \\
 & y_1 \leq 30\,000Z \\
 & y_2 \leq 30\,000(1 - Z) \\
 & 10\,000 - k \leq 10\,000Z \\
 & x_{ij} \geq 0 \quad \forall i \in \{1, 2\}, j \in \{1, 2\} \\
 & y_i \geq 0 \quad \forall i \in \{1, 2\} \\
 & Z \in \{0, 1\}
 \end{aligned}$$

option 2: Alternatively, the costs can be introduced as an additional variable $c \geq 0$ that needs to be optimized. The two variants are modeled as different lower limits for c . The upper of the two limits (= no discount) only has to apply if $k \leq 10\,000$. For this, we introduce the variable $Z \in \{0, 1\}$, for which the following applies: $k < 10\,000 \rightarrow Z = 0$.

$$\begin{array}{llll}
\max z = & 1.4x_{11} + 1.4x_{21} + 1.5x_{12} + 1.5x_{22} - c & & \\
\text{s.t.} & x_{11} + x_{12} & \leq & 10\,000 + k \\
& x_{21} + x_{22} & \leq & 20\,000 \\
& 0.5x_{21} - 0.5x_{11} & \leq & 0 \\
& 0.6x_{22} - 0.4x_{12} & \leq & 0 \\
& k & \leq & 30\,000 \\
& 10\,000Z & \leq & k \\
& c & \geq & 1.8(k - M \cdot Z) \\
& c & \geq & 1.6(k - M \cdot (1 - Z)) \\
& x_{ij} & \geq & 0 \qquad \forall i \in \{1, 2\}, j \in \{1, 2\} \\
& c & \geq & 0 \\
& Z & \in & \{0, 1\}
\end{array}$$